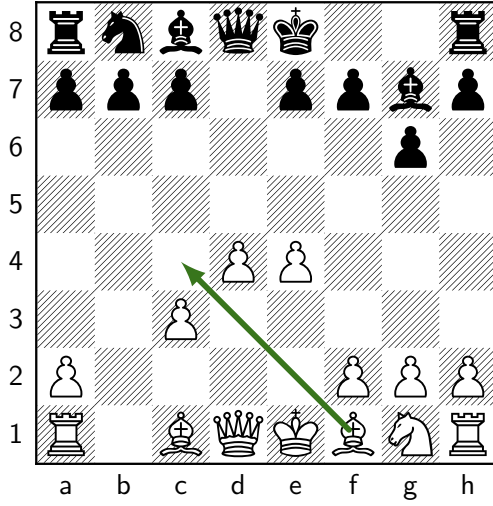


Melanie-BOTS (CCRL engines) — Grunfeld

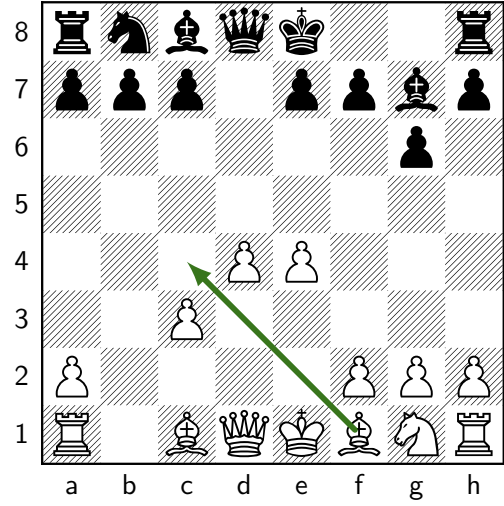
Melanie (Deep_Vision)

Abstract

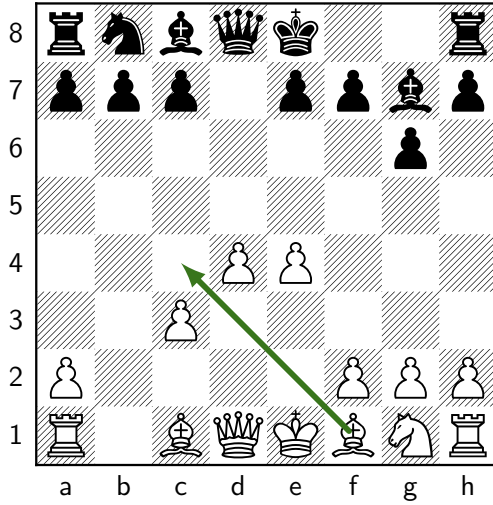
Every move below was produced by relational analogy to named master games (human and/or top-engine – the exact source is named per move). Each figure shows our position with Melanie’s move (a) and the top retrieved analog positions (b–d) with the master move played there; the caption names the exact source and the piece-role correspondence under μ – cross-type maps (a rook playing a queen’s role) are called out, as no material/eval engine explains a move that way.



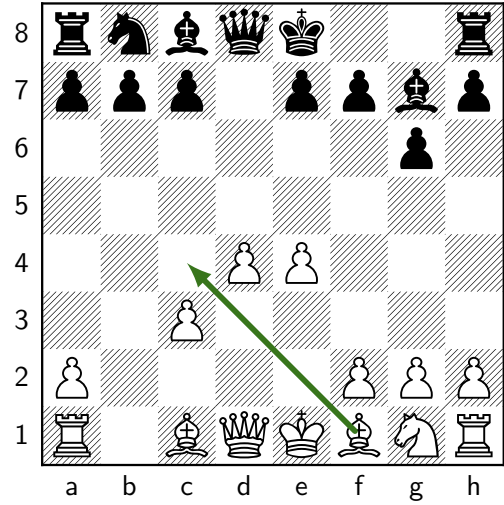
(a)



(b)

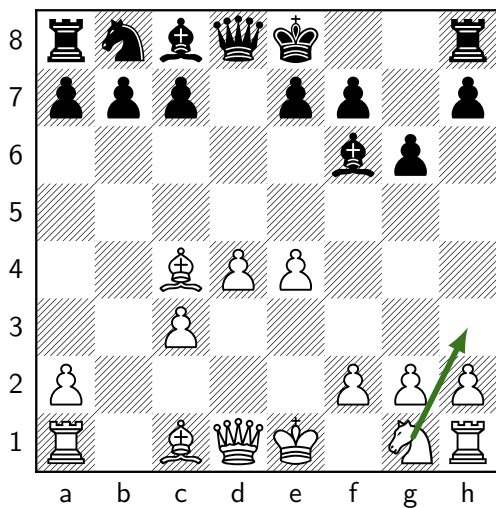


(c)

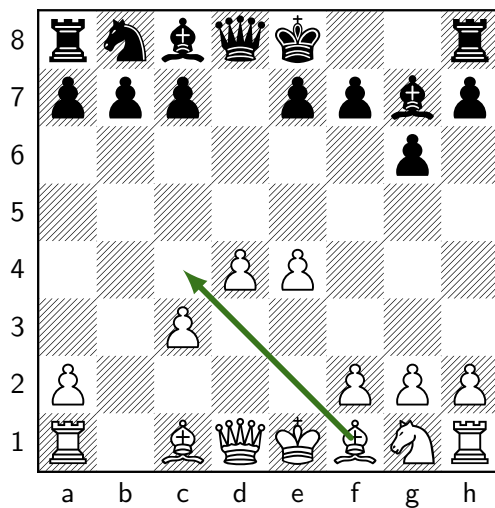


(d)

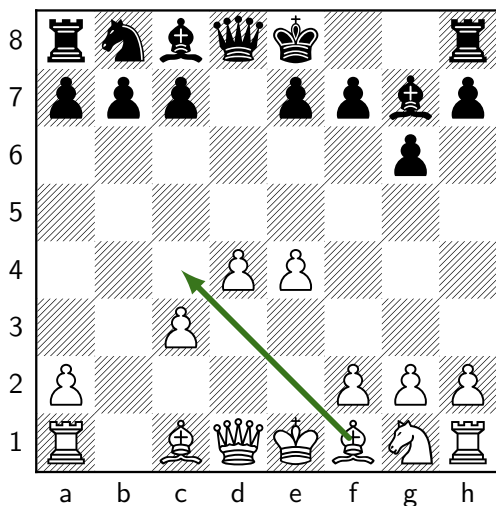
Figure 1: **Move 7, White: Bc4.** (a) Our position; Melanie plays Bc4 (f1c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#366` (ply 12) [structural distance $d=0$, QAP obj=87]: the master played Bc4; under μ our Bf1 plays the role of that game’s Bf1; cross-type roles: our pg6 as their Bc1, our Bc1 as their Ng1; transferred move f1c4 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#367` (ply 12) [structural distance $d=0$, QAP obj=87]: the master played Bc4; under μ our Bf1 plays the role of that game’s Bf1; cross-type roles: our pg6 as their Bc1, our Bc1 as their Ng1; transferred move f1c4 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#440` (ply 12) [structural distance $d=0$, QAP obj=87]: the master played Bc4; under μ our Bf1 plays the role of that game’s Bf1; cross-type roles: our pg6 as their Bc1, our Bc1 as their Ng1; transferred move f1c4 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Bc4 (summed QAP objective 1043, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



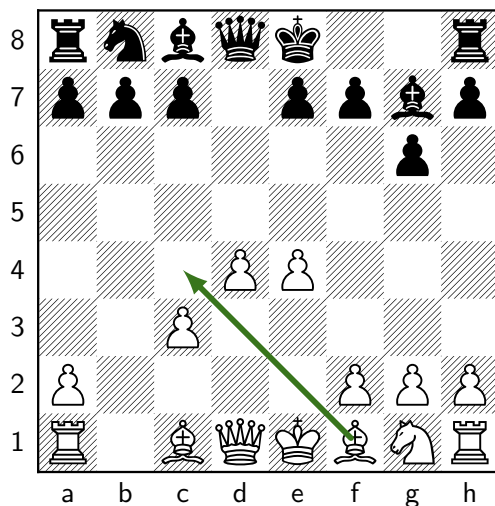
(a)



(b)

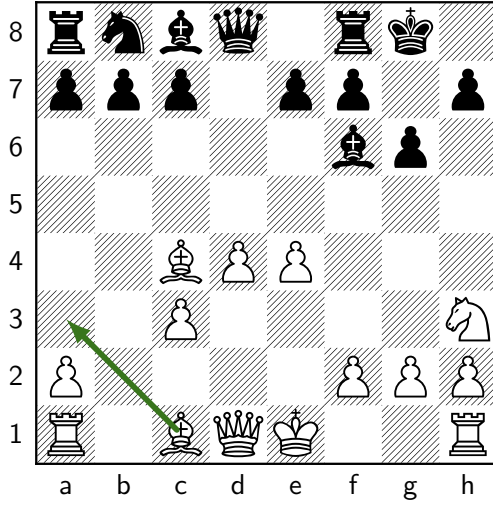


(c)

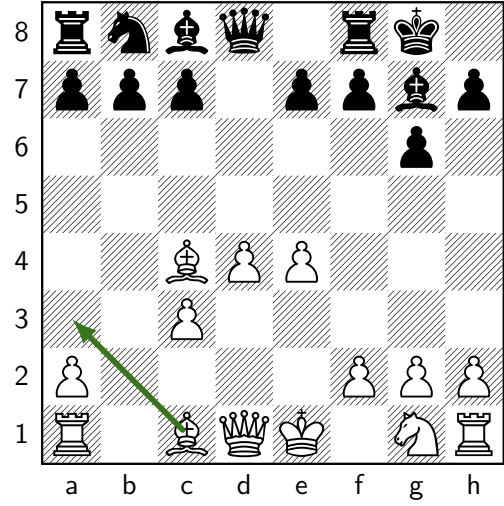


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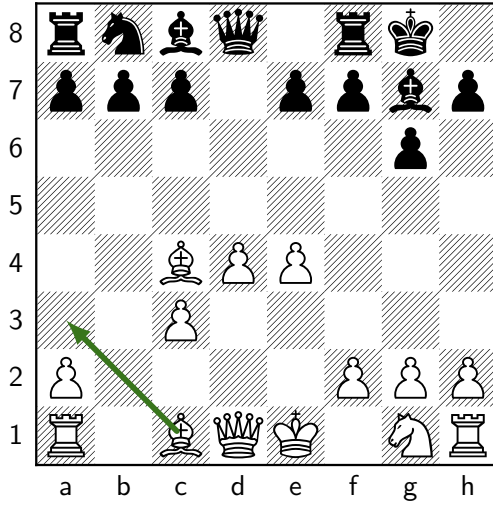
Figure 2: **Move 8, White: Nh3.** (a) Our position; Melanie plays Nh3 (g1h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#366 (ply 12) [structural distance $d=19$, QAP obj=78]: the master played Bc4; under μ our Ng1 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Ph2 as their Ng1, our Bc4 as their pe7; transferred move g1h3 (**endorses this move**). (c) Retrieved from ccr1_top_p0000#367 (ply 12) [structural distance $d=19$, QAP obj=78]: the master played Bc4; under μ our Ng1 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Ph2 as their Ng1, our Bc4 as their pe7; transferred move g1h3 (**endorses this move**). (d) Retrieved from ccr1_top_p0000#440 (ply 12) [structural distance $d=19$, QAP obj=78]: the master played Bc4; under μ our Ng1 plays the role of that game’s Bf1 (a **cross-type** correspondence); cross-type roles: our Ph2 as their Ng1, our Bc4 as their pe7; transferred move g1h3 (**endorses this move**). Synthesis: 12 retrieved analogs transferred to Nh3 (summed QAP objective 940, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



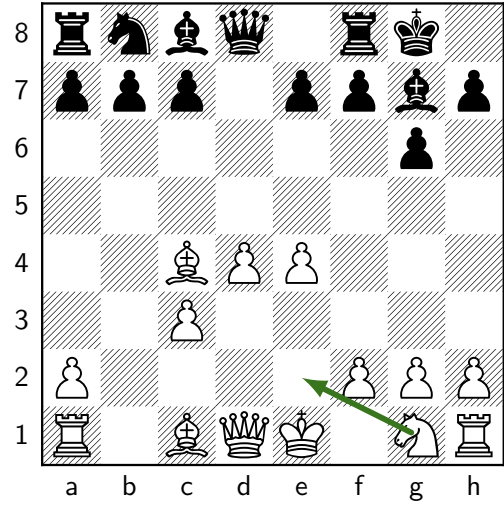
(a)



(b)

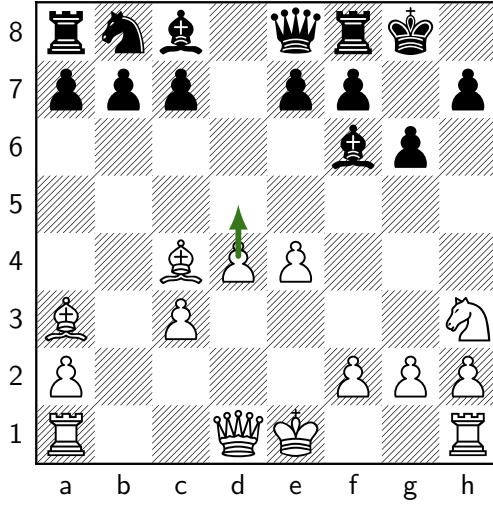


(c)

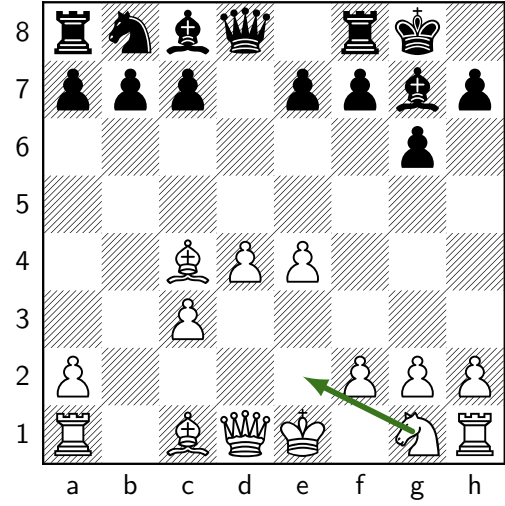


(d)

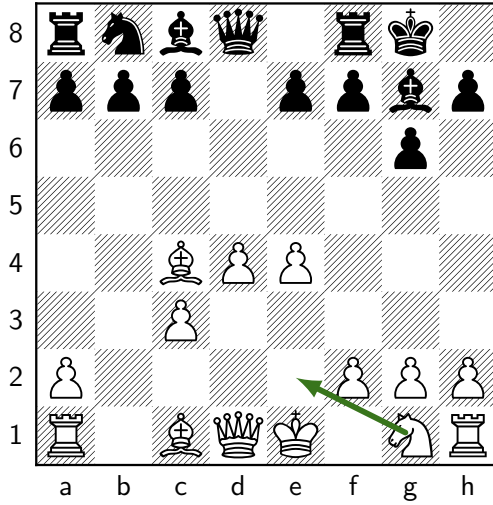
Figure 3: **Move 9, White: Ba3.** (a) Our position; Melanie plays Ba3 (c1a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `cctl_top_p0000#3872` (ply 14) [structural distance $d=23$, QAP obj=109]: the master played Ba3; under μ our Bc1 plays the role of that game’s Bc1; cross-type roles: our Pg2 as their Ng1, our Nh3 as their Ph2; transferred move c1a3 (**endorses this move**). (c) Retrieved from `cctl_top_p0000#3873` (ply 14) [structural distance $d=23$, QAP obj=109]: the master played Ba3; under μ our Bc1 plays the role of that game’s Bc1; cross-type roles: our Pg2 as their Ng1, our Nh3 as their Ph2; transferred move c1a3 (**endorses this move**). (d) Retrieved from `cctl_top_p0000#440` (ply 14) [structural distance $d=23$, QAP obj=109]: the master played Ne2; under μ our Pg2 plays the role of that game’s Ng1 (a **cross-type** correspondence); cross-type roles: our Nh3 as their Ph2; transferred move g2g3. Synthesis: 2 retrieved analogs transferred to Ba3 (summed QAP objective 218, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



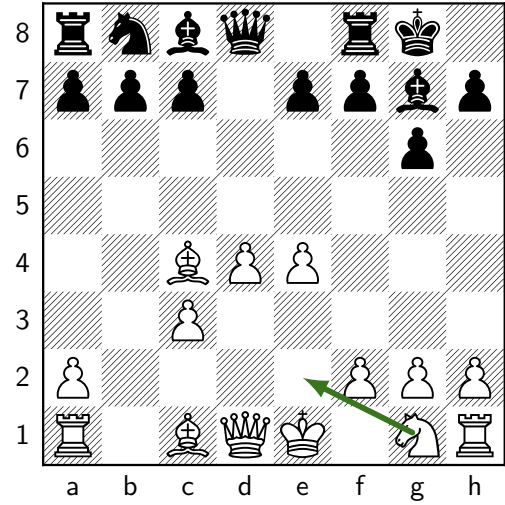
(a)



(b)

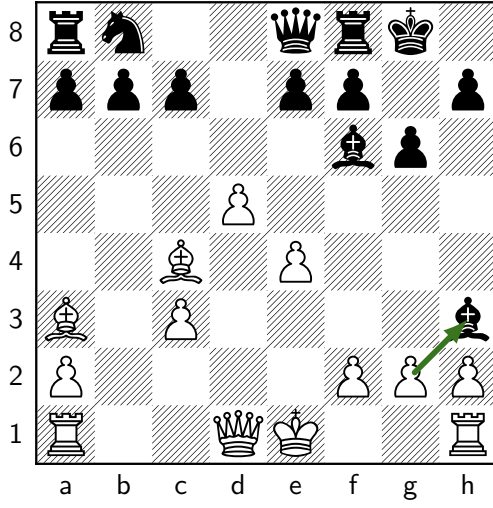


(c)

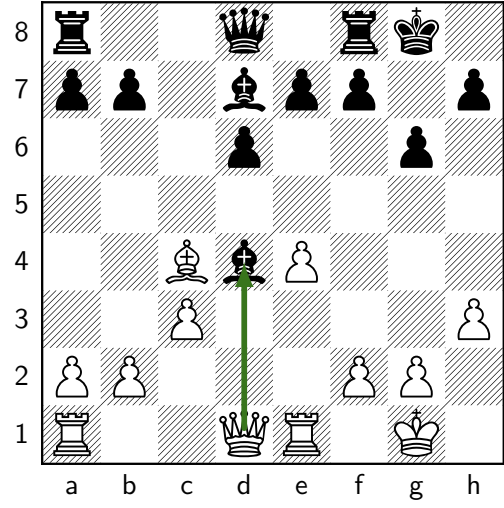


(d)

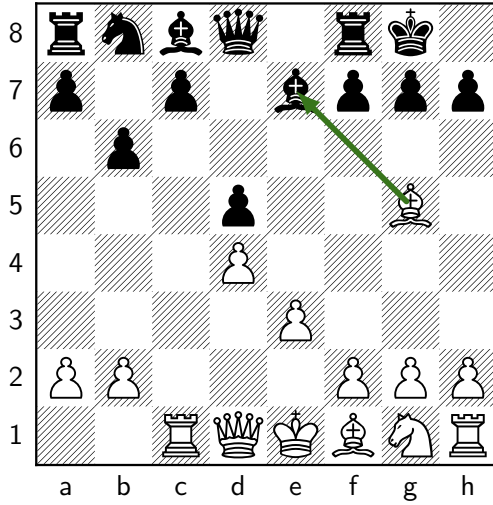
Figure 4: **Move 10, White: d5.** (a) Our position; Melanie plays d5 (d4d5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#440` (ply 14) [structural distance $d=44$, QAP obj=72]: the master played $Ne2$; under μ our $Pd4$ plays the role of that game’s $Ng1$ (a **cross-type** correspondence); cross-type roles: our $bc8$ as their $Ra1$, our $pc7$ as their $Bc1$; transferred move d4d5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#441` (ply 14) [structural distance $d=44$, QAP obj=72]: the master played $Ne2$; under μ our $Pd4$ plays the role of that game’s $Ng1$ (a **cross-type** correspondence); cross-type roles: our $bc8$ as their $Ra1$, our $pc7$ as their $Bc1$; transferred move d4d5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#1228` (ply 14) [structural distance $d=44$, QAP obj=72]: the master played $Ne2$; under μ our $Pd4$ plays the role of that game’s $Ng1$ (a **cross-type** correspondence); cross-type roles: our $bc8$ as their $Ra1$, our $pc7$ as their $Bc1$; transferred move d4d5 (**endorses this move**). Synthesis: 14 retrieved analogs transferred to d5 (summed QAP objective 1003, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



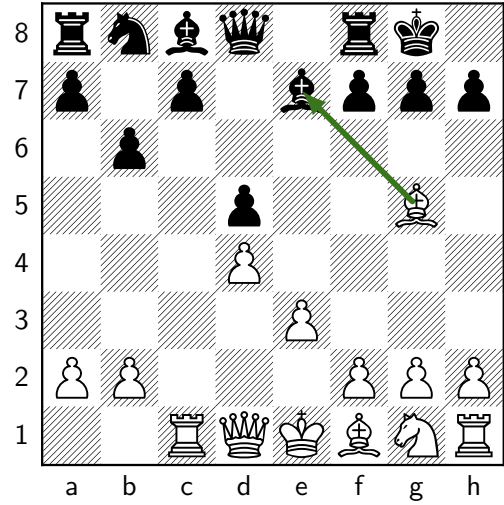
(a)



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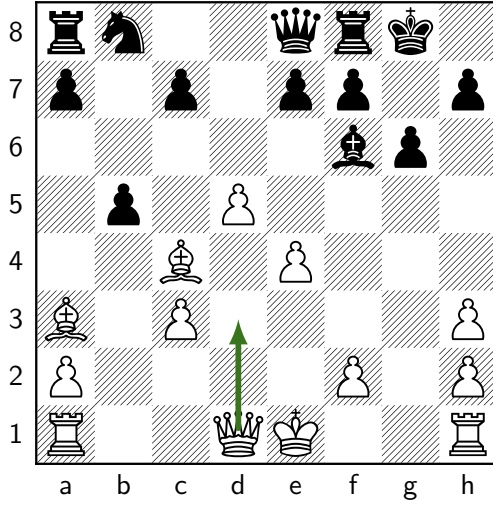


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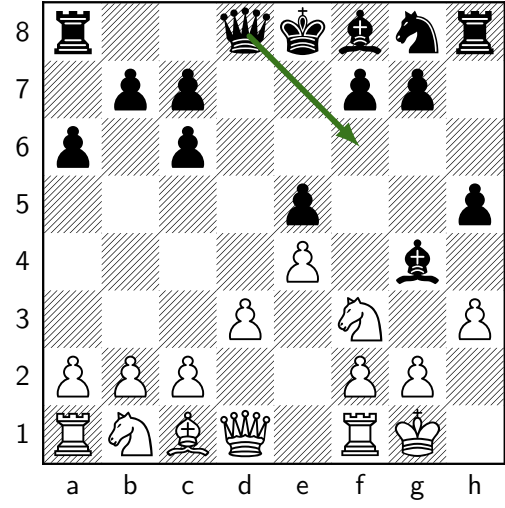


(d)

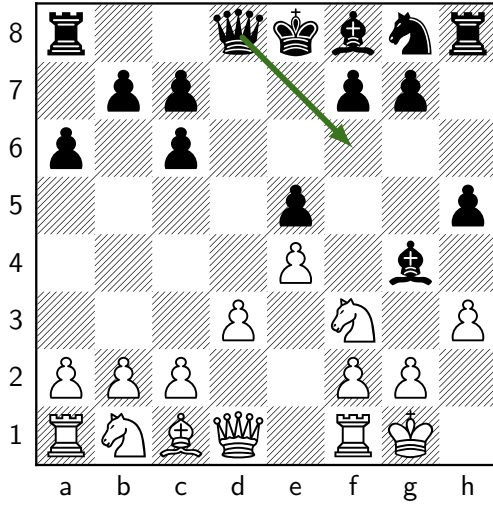
Figure 5: **Move 11, White: gxh3.** (a) Our position; Melanie plays gxh3 (g2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0002#280 (ply 28) [structural distance $d=59$, QAP obj=50]: the master played Qxd4; under μ our Pg2 plays the role of that game’s Qd1 (a **cross-type** correspondence); cross-type roles: our pc7 as their Ra1, our Qd1 as their Re1; transferred move g2h3 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#3321 (ply 16) [structural distance $d=55$, QAP obj=22]: the master played Bxe7; under μ our Pg2 plays the role of that game’s Bg5 (a **cross-type** correspondence); cross-type roles: our qe8 as their Rc1, our pf7 as their Bf1; transferred move g2g4. (d) Retrieved from ccr1_top_p0001#3322 (ply 16) [structural distance $d=55$, QAP obj=22]: the master played Bxe7; under μ our Pg2 plays the role of that game’s Bg5 (a **cross-type** correspondence); cross-type roles: our qe8 as their Rc1, our pf7 as their Bf1; transferred move g2g4. Synthesis: 1 retrieved analog transferred to gxh3 (summed QAP objective 50, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



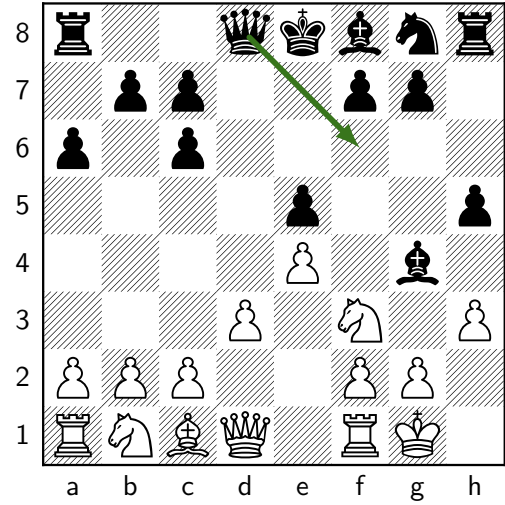
(a)



(b)

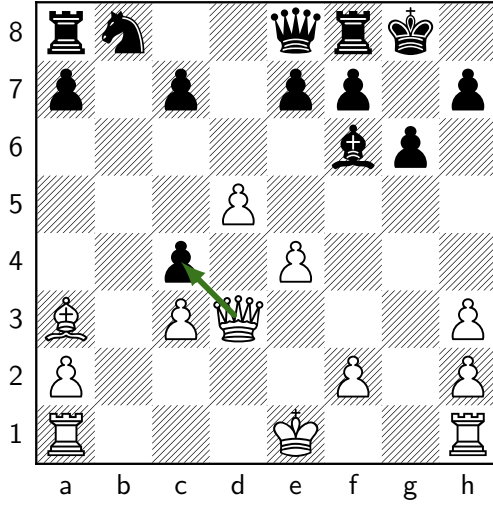


(c)

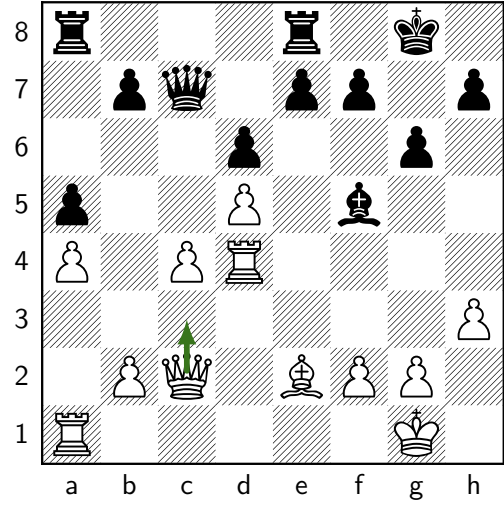


(d)

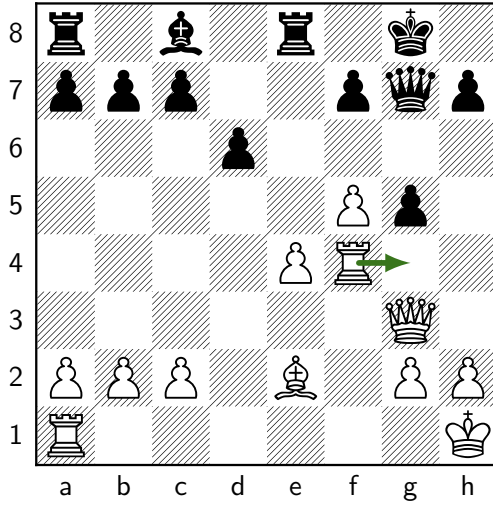
Figure 6: **Move 12, White: Qd3.** (a) Our position; Melanie plays Qd3 (d1d3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#2713` (ply 13) [structural distance $d=72$, QAP obj=30]: the master played Qf6; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our pf7 as their Ra8, our Ra1 as their Ng8; transferred move d1d3 (**endorses this move**). (c) Retrieved from `ccr1_top_p0002#2714` (ply 13) [structural distance $d=72$, QAP obj=30]: the master played Qf6; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our pf7 as their Ra8, our Ra1 as their Ng8; transferred move d1d3 (**endorses this move**). (d) Retrieved from `ccr1_top_p0003#1830` (ply 13) [structural distance $d=72$, QAP obj=30]: the master played Qf6; under μ our Qd1 plays the role of that game’s Qd1; cross-type roles: our pf7 as their Ra8, our Ra1 as their Ng8; transferred move d1d3 (**endorses this move**). Synthesis: 4 retrieved analogs transferred to Qd3 (summed QAP objective 121, the strongest correspondence among the candidate moves); Melanie committed it via the `search` selector.



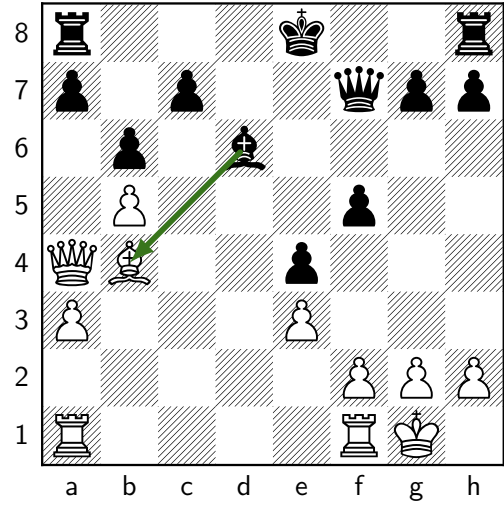
(a)



(b)

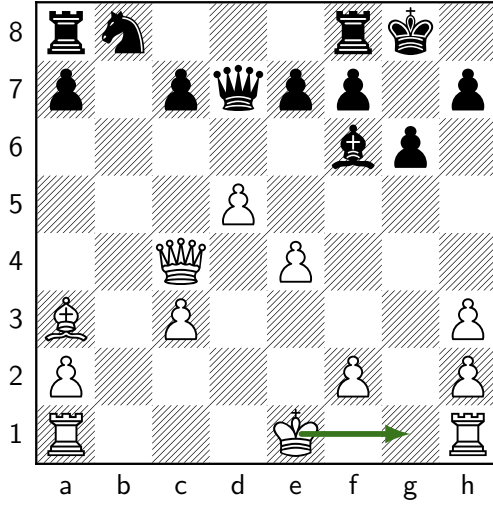


(c)

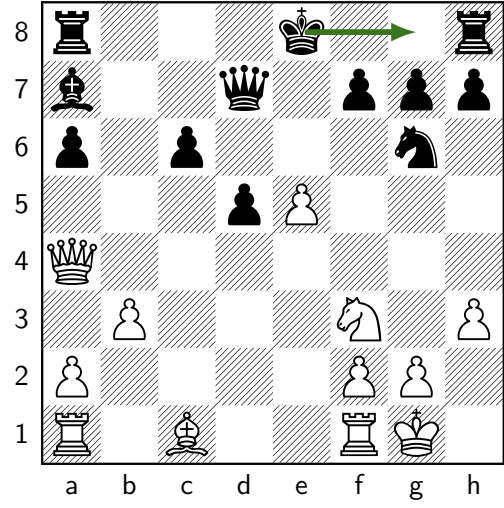


(d)

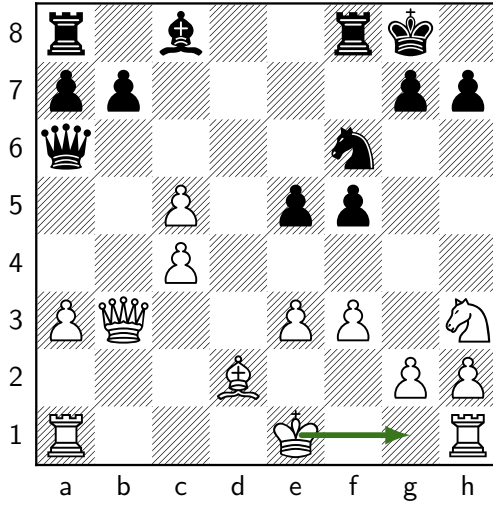
Figure 7: **Move 13, White: Qxc4.** (a) Our position; Melanie plays Qxc4 (d3c4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0000#1042` (ply 34) [structural distance $d=72$, QAP obj=35]: the master played Qc3; under μ our Qd3 plays the role of that game’s Qc2; cross-type roles: our Rh1 as their Pb2, our Pd5 as their Be2; transferred move d3e3. (c) Retrieved from `ccrl_top_p0003#2435` (ply 34) [structural distance $d=74$, QAP obj=33]: the master played Rg4; under μ our Qd3 plays the role of that game’s Rf4 (a **cross-type** correspondence); cross-type roles: our Rh1 as their Pa2, our Pd5 as their Be2; transferred move d3g3. (d) Retrieved from `ccrl_top_p0001#2471` (ply 41) [structural distance $d=68$, QAP obj=30]: the master played Bxb4; under μ our Pd5 plays the role of that game’s Bd3 (a **cross-type** correspondence); cross-type roles: our Qd3 as their Ra8, our Pf2 as their Qf7; transferred move d5d6. Synthesis: no retrieved analog transferred to Qxc4 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



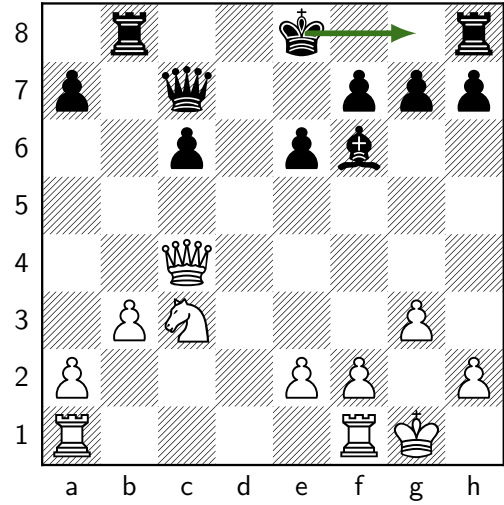
(a)



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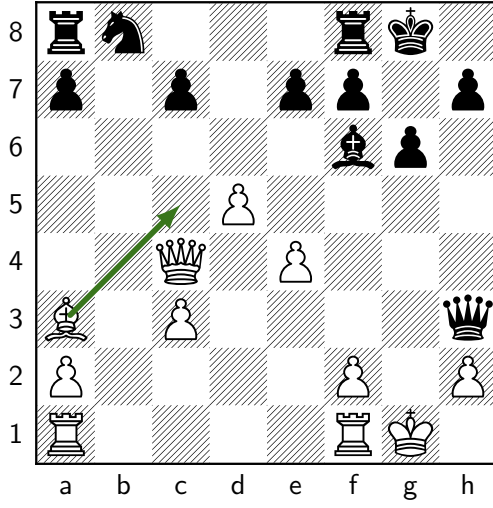


(c)

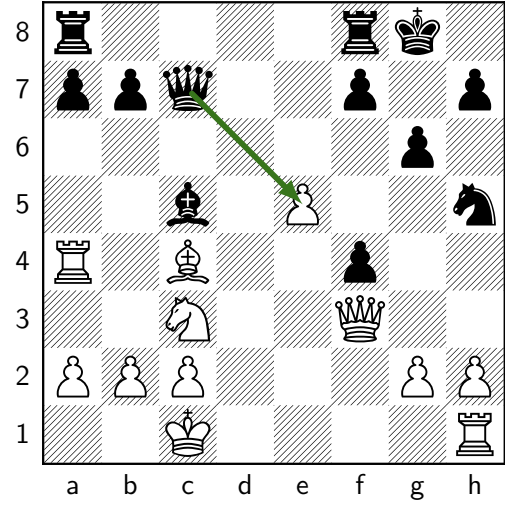


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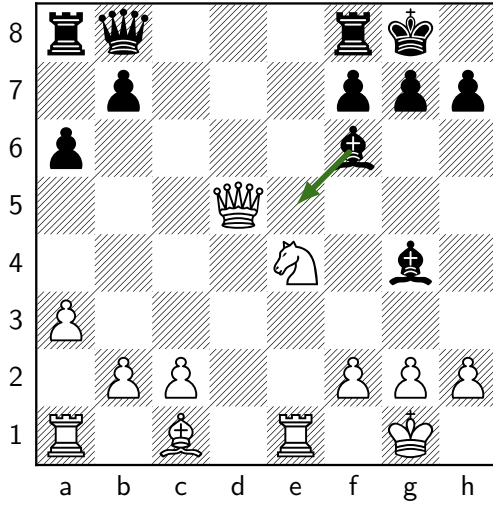
Figure 8: **Move 14, White: 0-0.** (a) Our position; Melanie plays 0-0 (**e1g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#436` (ply 35) [structural distance $d=82$, QAP obj=96]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *qd7* as their *Ng6*, our *bf6* as their *qa4*; transferred move **e1g1 (endorses this move)**. (c) Retrieved from `ccr1_top_p0000#768` (ply 28) [structural distance $d=65$, QAP obj=59]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Pe4* as their *Nh3*, our *qd7* as their *pe5*; transferred move **e1g1 (endorses this move)**. (d) Retrieved from `ccr1_top_p0003#821` (ply 31) [structural distance $d=81$, QAP obj=41]: the master played 0-0; under μ our *Ke1* plays the role of that game’s *Ke1*; cross-type roles: our *Ba3* as their *Rb8*, our *Pc3* as their *nc3*; transferred move **e1g1 (endorses this move)**. Synthesis: 4 retrieved analogs transferred to 0-0 (summed QAP objective 231, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



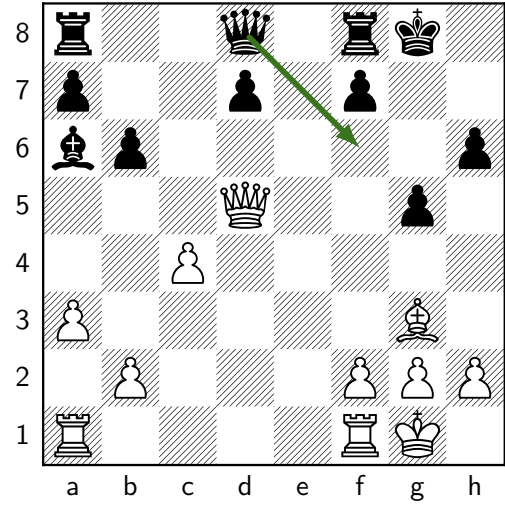
(a)



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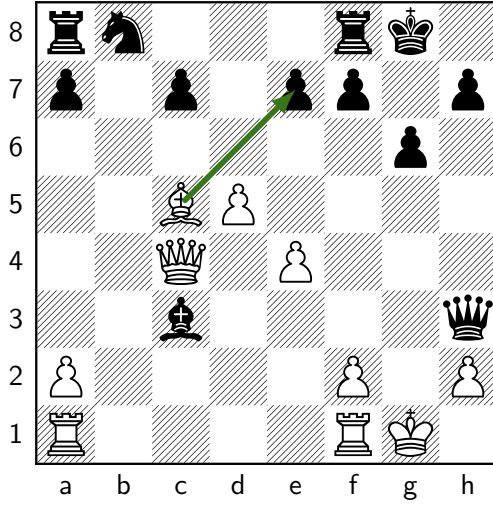


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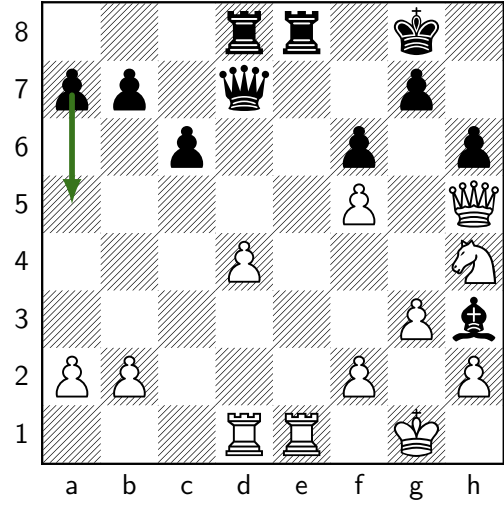


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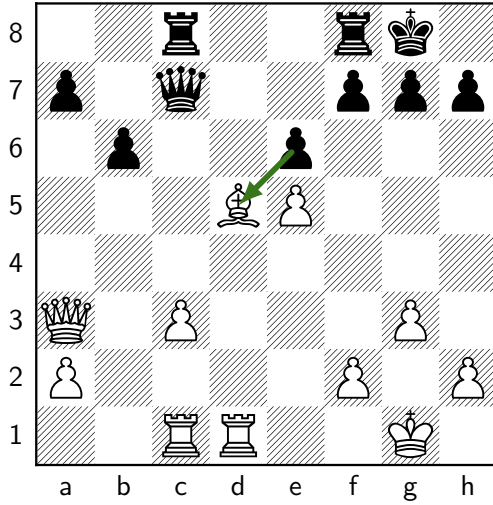
Figure 9: **Move 15, White: Bc5.** (a) Our position; Melanie plays Bc5 (a3c5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1363` (ply 35) [structural distance $d=75$, QAP obj=70]: the master played Qxe5; under μ our *Ba3* plays the role of that game’s *Qc2* (a **cross-type** correspondence); cross-type roles: our *Qc4* as their *Ra8*, our *Pe4* as their *Rf8*; transferred move a3c5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2967` (ply 39) [structural distance $d=70$, QAP obj=40]: the master played Be5; under μ our *Ba3* plays the role of that game’s *Bf3*; cross-type roles: our *nb8* as their *Ra8*, our *bf6* as their *qd5*; transferred move a3c5 (**endorses this move**). (d) Retrieved from `ccr1_top_p0000#3972` (ply 33) [structural distance $d=75$, QAP obj=73]: the master played Qf6; under μ our *Qc4* plays the role of that game’s *Qd1*; cross-type roles: our *Pe4* as their *Rf8*, our *Rf1* as their *Pg5*; transferred move c4d3. Synthesis: 2 retrieved analogs transferred to Bc5 (summed QAP objective 110, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



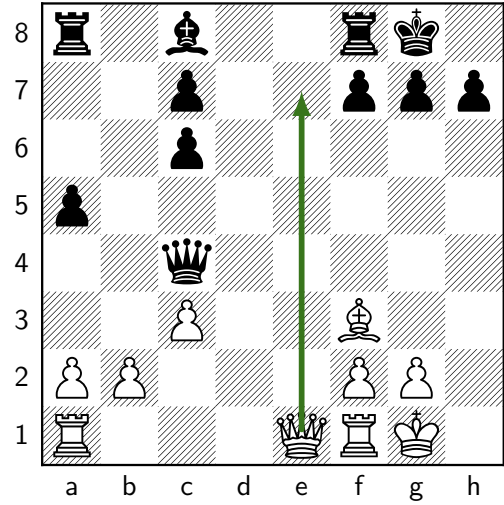
(a)



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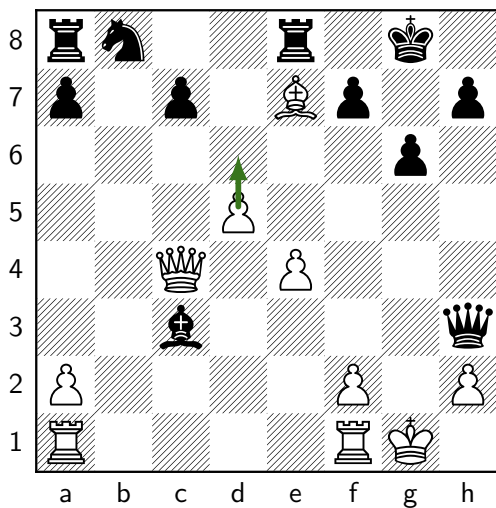


(c)

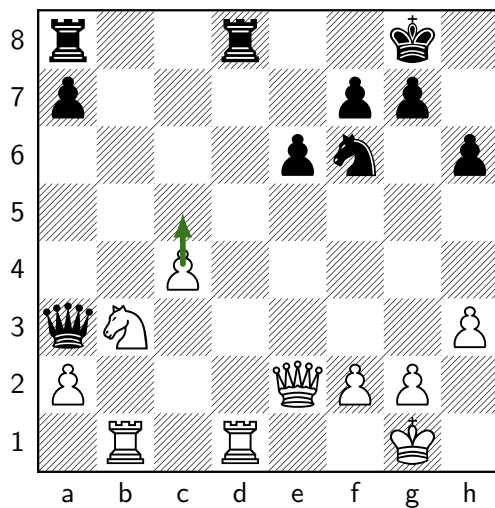


(d)

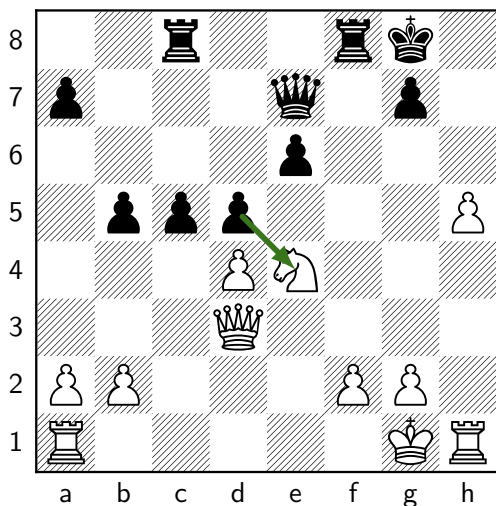
Figure 10: **Move 16, White: Bxe7.** (a) Our position; Melanie plays **Bxe7** (**c5e7**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#3461` (ply 43) [structural distance $d=72$, QAP obj=46]: the master played **a5**; under μ our *Pf2* plays the role of that game’s *Pa2*; cross-type roles: our *Qc4* as their *Rd8*, our *Bc5* as their *Qd7*; transferred move **f2f4**. (c) Retrieved from `ccrl_top_p0000#2433` (ply 37) [structural distance $d=66$, QAP obj=46]: the master played **exd5**; under μ our *Qc4* plays the role of that game’s *Pe3* (a **cross-type** correspondence); cross-type roles: our *Pd5* as their *Rc8*, our *Bc5* as their *Qc7*; transferred move **c4d3**. (d) Retrieved from `ccrl_top_p0001#2859` (ply 38) [structural distance $d=68$, QAP obj=42]: the master played **Qe7**; under μ our *Ra1* plays the role of that game’s *Qe1* (a **cross-type** correspondence); cross-type roles: our *Qc4* as their *Ra1*, our *ra8* as their *pc7*; transferred move **a1c1**. Synthesis: no retrieved analog transferred to **Bxe7** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



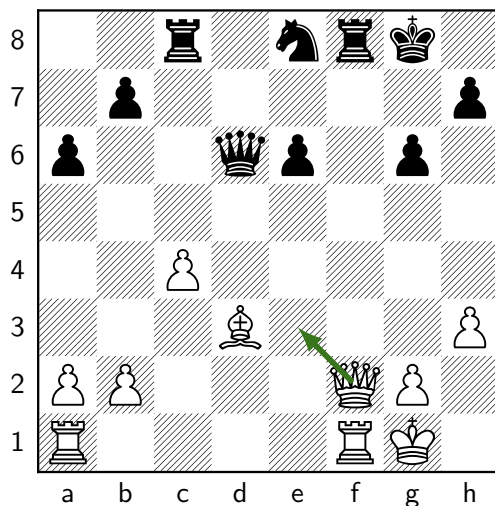
(a)



(b)

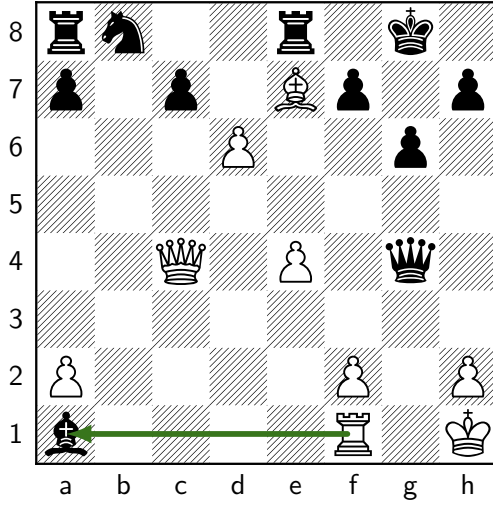


(c)

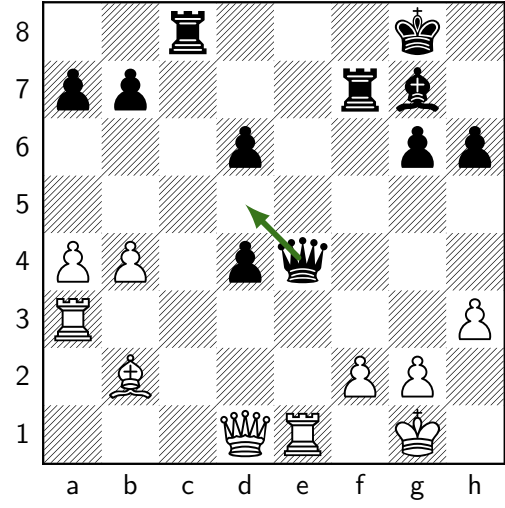


(d)

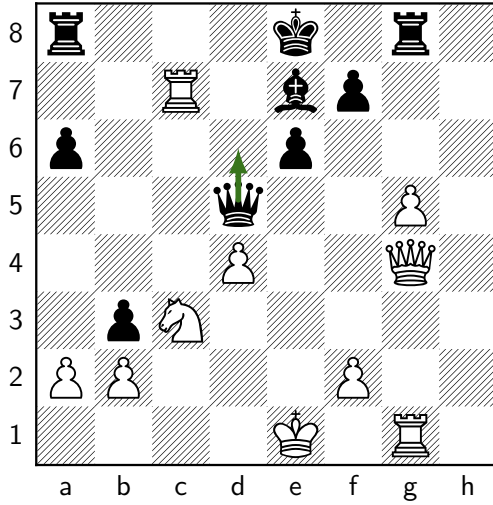
Figure 11: **Move 17, White: d6.** (a) Our position; Melanie plays d6 (d5d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0002#3764` (ply 46) [structural distance $d=66$, QAP obj=34]: the master played c5; under μ our *Pd5* plays the role of that game’s *Pc4*; cross-type roles: our *Be7* as their *Rd1*, our *Rf1* as their *Nb3*; transferred move d5d6 (**endorses this move**). (c) Retrieved from `ccrl_top_p0003#1917` (ply 45) [structural distance $d=67$, QAP obj=32]: the master played dxe4; under μ our *Qc4* plays the role of that game’s *Pd4* (a **cross-type** correspondence); cross-type roles: our *pc7* as their *Rf8*, our *pf7* as their *Qe7*; transferred move c4d3. (d) Retrieved from `ccrl_top_p0003#739` (ply 46) [structural distance $d=68$, QAP obj=23]: the master played Qe3; under μ our *Be7* plays the role of that game’s *Qf2* (a **cross-type** correspondence); cross-type roles: our *qh3* as their *Ph3*, our *nb8* as their *Pc4*; transferred move e7c5. Synthesis: 1 retrieved analog transferred to d6 (summed QAP objective 34, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



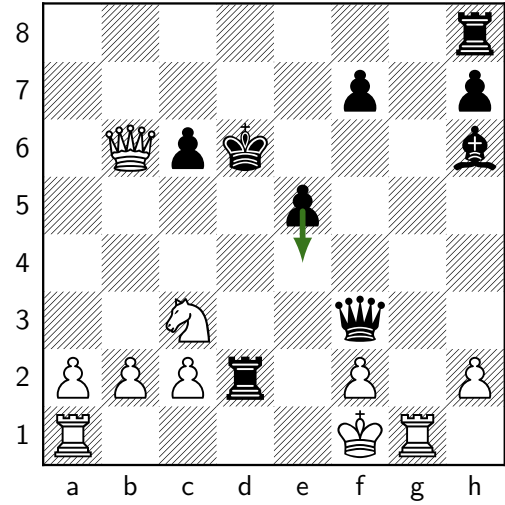
(a)



(b)

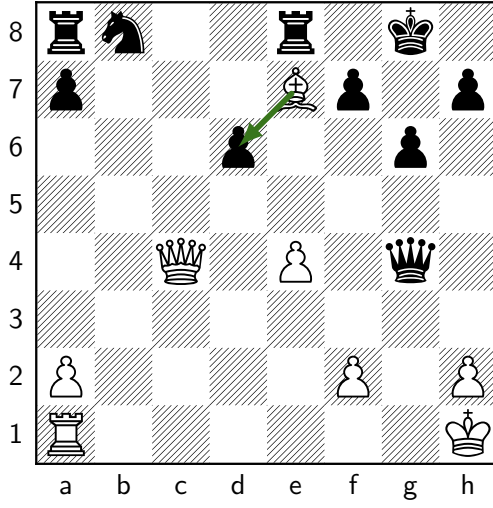


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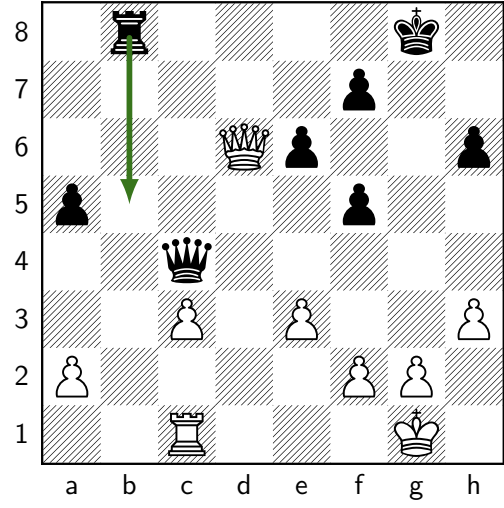


(d)

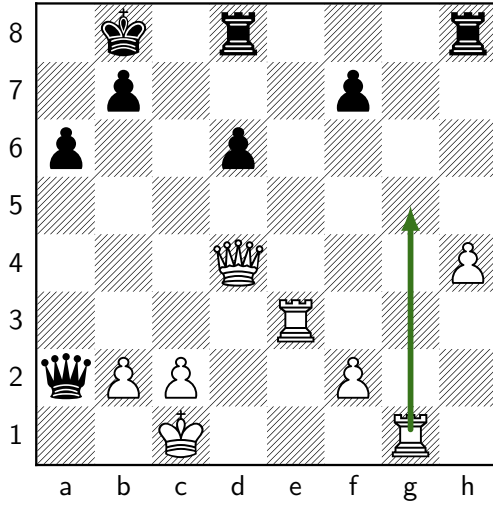
Figure 12: **Move 19, White: Rxa1**. (a) Our position; Melanie plays Rxa1 (f1a1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0003#2890 (ply 45) [structural distance $d=72$, QAP obj=42]: the master played Qd5; under μ our *Be7* plays the role of that game’s *Qe5* (a **cross-type** correspondence); cross-type roles: our *ph7* as their *Rc8*, our *ba1* as their *Pa7*; transferred move e7f6. (c) Retrieved from ccr1_top_p0001#250 (ply 43) [structural distance $d=71$, QAP obj=39]: the master played Qd6; under μ our *Be7* plays the role of that game’s *Qd4* (a **cross-type** correspondence); cross-type roles: our *gg4* as their *Rg8*, our *pf7* as their *rc7*; transferred move e7f6. (d) Retrieved from ccr1_top_p0002#1650 (ply 43) [structural distance $d=60$, QAP obj=37]: the master played e4; under μ our *Ph2* plays the role of that game’s *Pe4*; cross-type roles: our *pc7* as their *nc3*, our *Pd6* as their *Rd2*; transferred move h2h4. Synthesis: no retrieved analog transferred to Rxa1 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



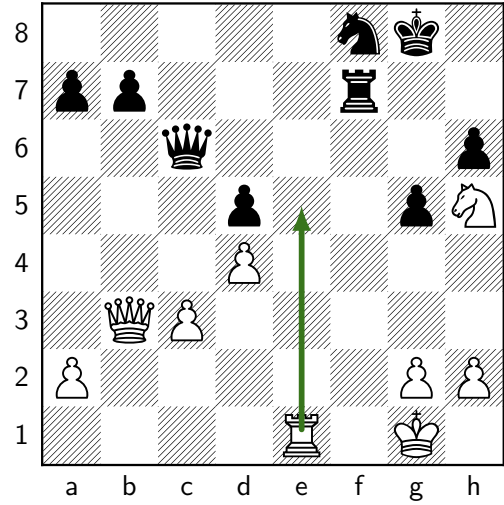
(a)



(b)

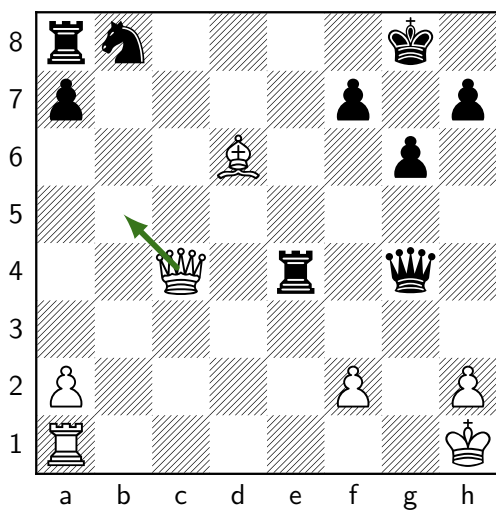


(c)

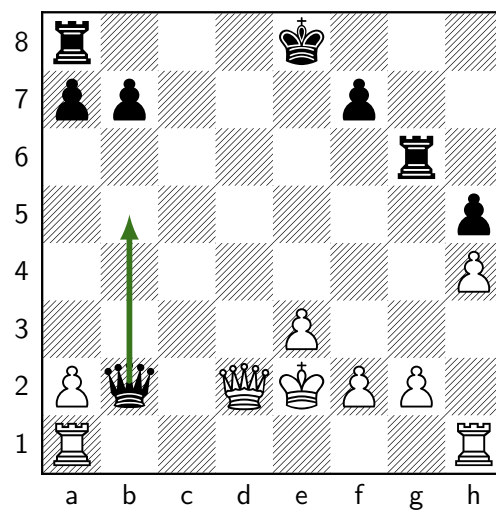


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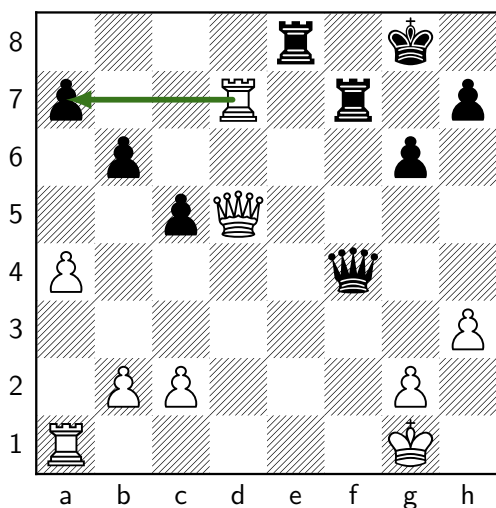
Figure 13: **Move 20, White: Bxd6.** (a) Our position; Melanie plays Bxd6 (e7d6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#381` (ply 53) [structural distance $d=60$, QAP obj=30]: the master played Rb5; under μ our *Be7* plays the role of that game’s *Rb1* (a **cross-type** correspondence); cross-type roles: our *re8* as their *qd6*, our *Be7* as their *Pe6*; transferred move e7h4. (c) Retrieved from `ccr1_top_p0001#16` (ply 52) [structural distance $d=63$, QAP obj=30]: the master played Rg5; under μ our *Ra1* plays the role of that game’s *Rg1*; cross-type roles: our *Be7* as their *qa2*, our *pg6* as their *Re3*; transferred move a1f1. (d) Retrieved from `ccr1_top_p0000#3342` (ply 46) [structural distance $d=58$, QAP obj=27]: the master played Re5; under μ our *Ra1* plays the role of that game’s *Re1*; cross-type roles: our *nb8* as their *Pd4*, our *pa7* as their *Nh5*; transferred move a1d1. Synthesis: no retrieved analog transferred to Bxd6 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



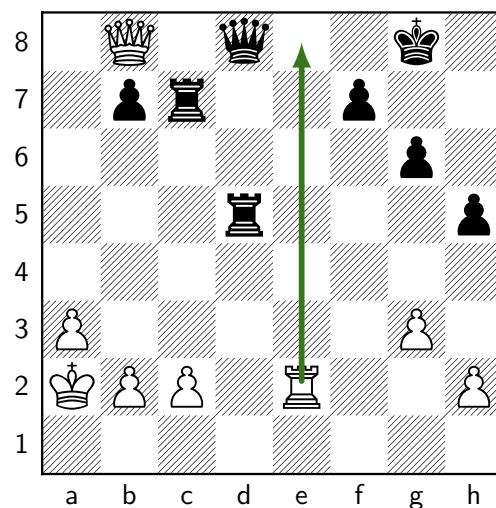
(a)



(b)

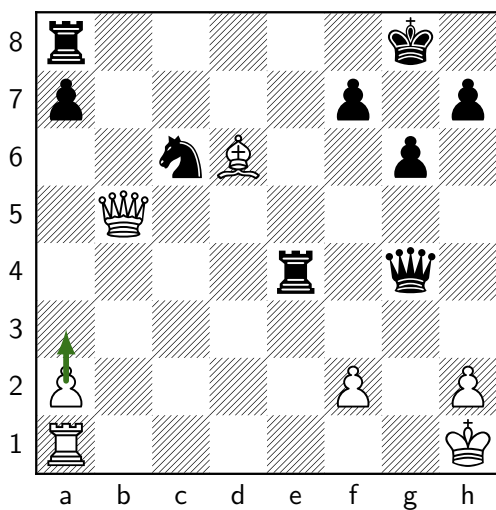


(c)

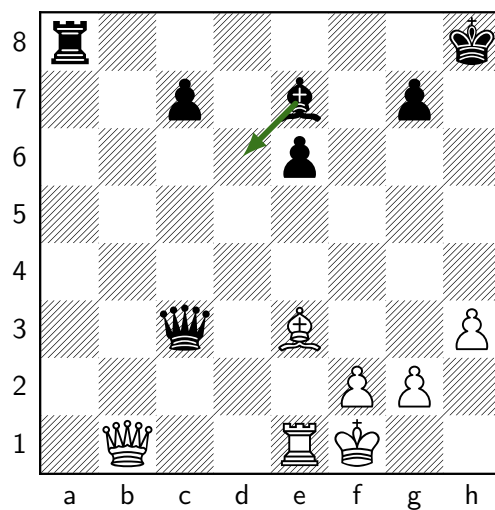


(d)

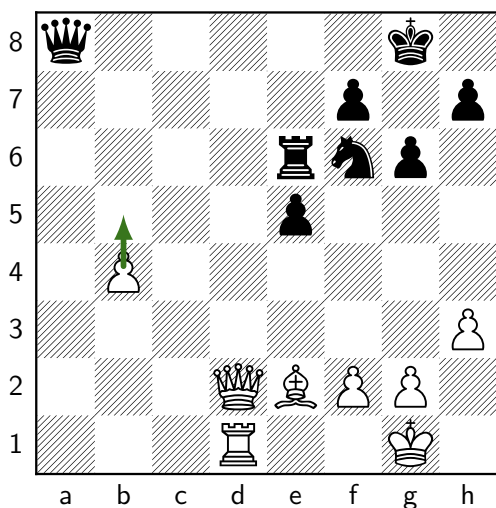
Figure 14: **Move 21, White: Qb5.** (a) Our position; Melanie plays Qb5 (c4b5). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#855` (ply 41) [structural distance $d=54$, QAP obj=45]: the master played Qb5+; under μ our Qc4 plays the role of that game’s Qb7; cross-type roles: our qg4 as their Rg6, our ra8 as their ph4; transferred move c4b5 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#1147` (ply 50) [structural distance $d=63$, QAP obj=58]: the master played Rxa7; under μ our Qc4 plays the role of that game’s Rd7 (a **cross-type** correspondence); cross-type roles: our nb8 as their Pa4, our Bd6 as their pg6; transferred move c4a4. (d) Retrieved from `ccr1_top_p0001#220` (ply 56) [structural distance $d=62$, QAP obj=45]: the master played Re8+; under μ our Ra1 plays the role of that game’s Re2; cross-type roles: our ra8 as their Pb2, our Bd6 as their Pa3; transferred move a1d1. Synthesis: 2 retrieved analogs transferred to Qb5 (summed QAP objective 91, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



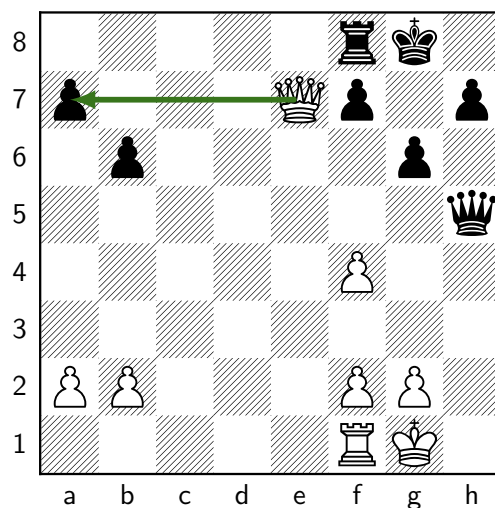
(a)



(b)

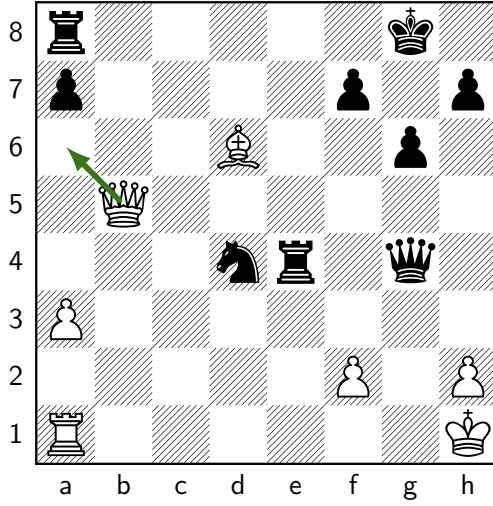


(c)

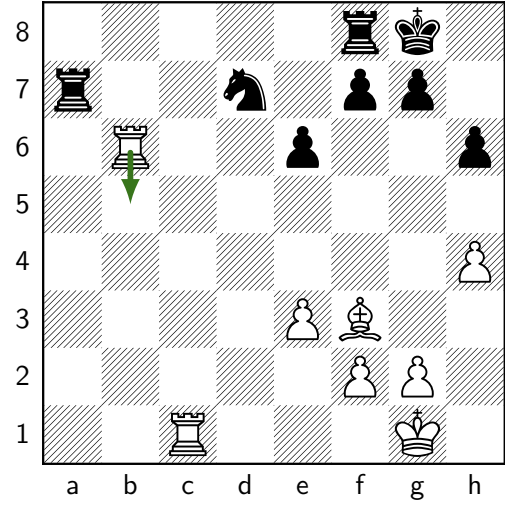


(d)

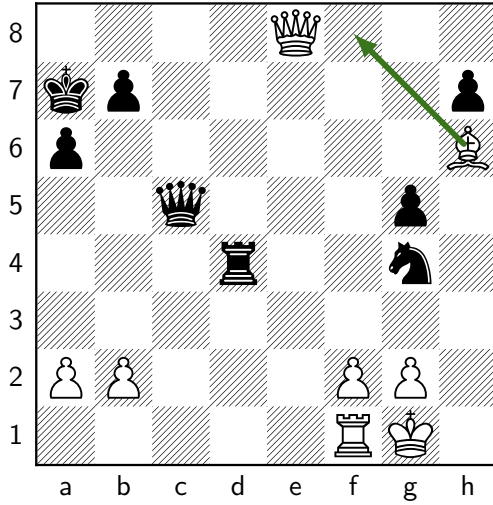
Figure 15: **Move 22, White: a3.** (a) Our position; Melanie plays a3 (a2a3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from ccr1_top_p0000#993 (ply 57) [structural distance $d=67$, QAP obj=22]: the master played Bd6; under μ our Pa2 plays the role of that game’s Be2 (a **cross-type** correspondence); cross-type roles: our Bd6 as their Pc7, our nc6 as their be3; transferred move a2a3 (**endorses this move**). (c) Retrieved from ccr1_top_p0001#642 (ply 48) [structural distance $d=66$, QAP obj=2]: the master played b5; under μ our Pa2 plays the role of that game’s Pb4; cross-type roles: our Ph2 as their Be2, our nc6 as their Pg2; transferred move a2a3 (**endorses this move**). (d) Retrieved from ccr1_top_p0002#3173 (ply 46) [structural distance $d=60$, QAP obj=13]: the master played Qxa7; under μ our Qb5 plays the role of that game’s Qe7; cross-type roles: our Bd6 as their Rf1, our re4 as their Pf2; transferred move b5a6. Synthesis: 2 retrieved analogs transferred to a3 (summed QAP objective 25, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



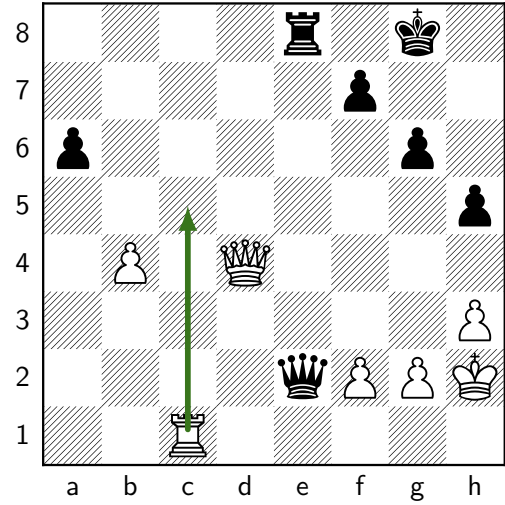
(a)



(b)

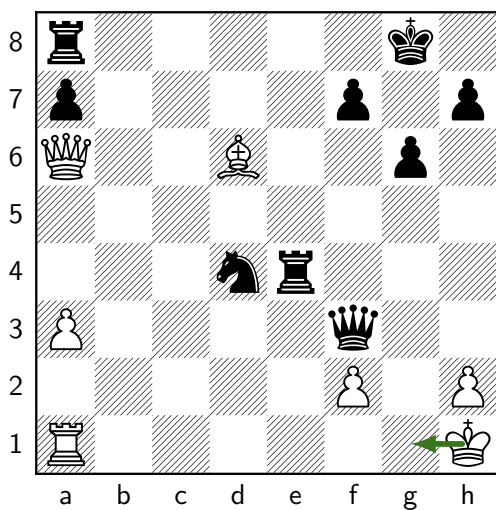


(c)

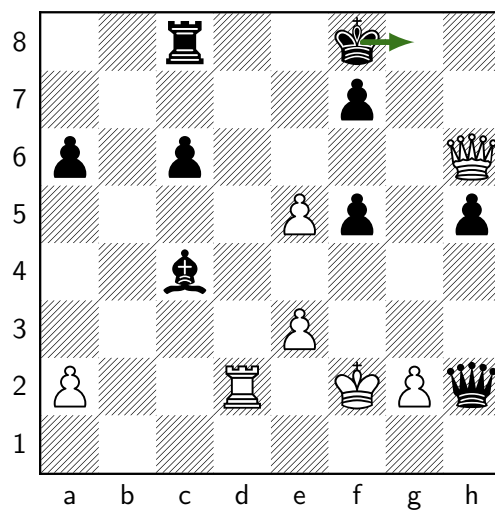


(d)

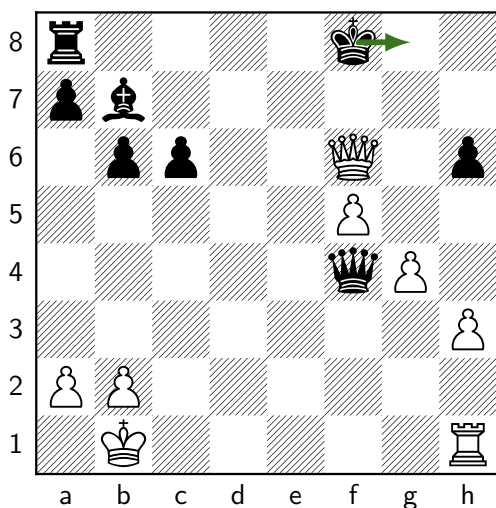
Figure 16: **Move 23, White: Qa6.** (a) Our position; Melanie plays Qa6 (b5a6). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0001#53` (ply 50) [structural distance $d=72$, QAP obj=12]: the master played Rb5; under μ our Qb5 plays the role of that game’s Rb6 (a **cross-type** correspondence); cross-type roles: our pg6 as their Bf3, our qg4 as their ph6; transferred move b5a5. (c) Retrieved from `ccr1_top_p0000#3093` (ply 54) [structural distance $d=69$, QAP obj=8]: the master played Bf8; under μ our Qb5 plays the role of that game’s Bh6 (a **cross-type** correspondence); cross-type roles: our ph7 as their Rf1, our Bd6 as their Pa2; transferred move b5b8. (d) Retrieved from `ccr1_top_p0000#2343` (ply 60) [structural distance $d=67$, QAP obj=5]: the master played Rc5; under μ our Ra1 plays the role of that game’s Rc1; cross-type roles: our nd4 as their qe2, our Qb5 as their Pf2; transferred move a1f1. Synthesis: no retrieved analog transferred to Qa6 – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



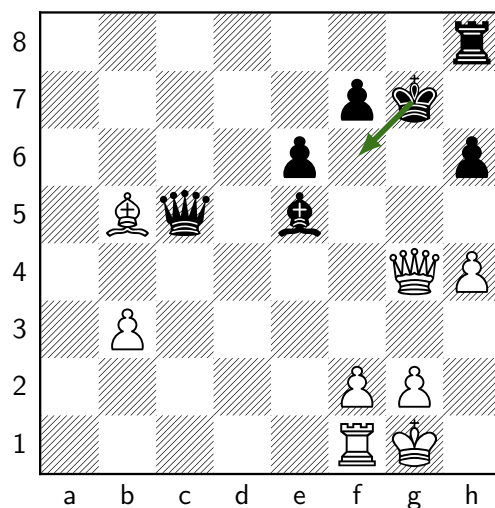
(a)



(b)

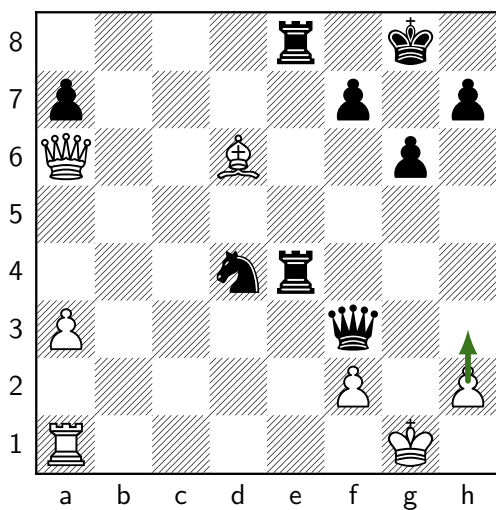


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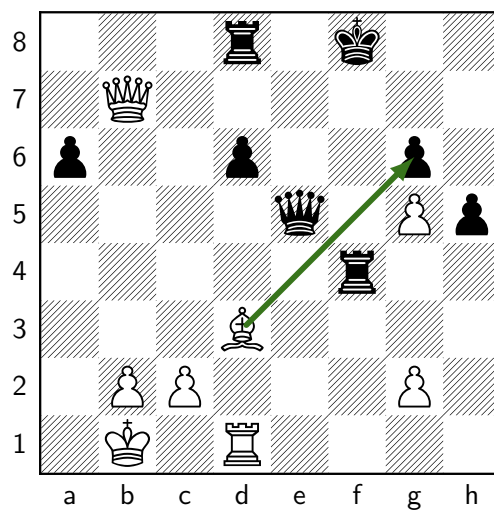


(d)

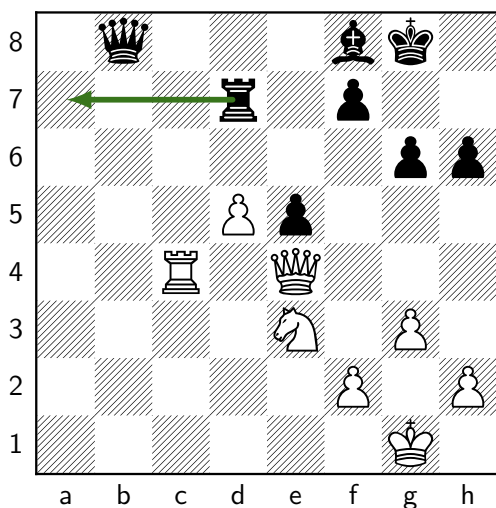
Figure 17: **Move 24, White: Kg1**. (a) Our position; Melanie plays Kg1 (**h1g1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0001#1068` (ply 61) [structural distance $d=51$, QAP obj=92]: the master played Kg8; under μ our *Kh1* plays the role of that game’s *Kf1*; cross-type roles: our *nd4* as their *pe5*, our *Bd6* as their *Ph5*; transferred move **h1g1 (endorses this move)**. (c) Retrieved from `ccrl_top_p0002#3220` (ply 53) [structural distance $d=41$, QAP obj=92]: the master played Kg8; under μ our *Kh1* plays the role of that game’s *Kf1*; cross-type roles: our *Pf2* as their *Bb7*, our *Bd6* as their *Ph6*; transferred move **h1g1 (endorses this move)**. (d) Retrieved from `ccrl_top_p0003#606` (ply 49) [structural distance $d=54$, QAP obj=57]: the master played Kf6; under μ our *Kh1* plays the role of that game’s *Kg2*; cross-type roles: our *Pa3* as their *Rh8*, our *nd4* as their *bb5*; transferred move **h1g1 (endorses this move)**. Synthesis: 7 retrieved analogs transferred to Kg1 (summed QAP objective 483, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



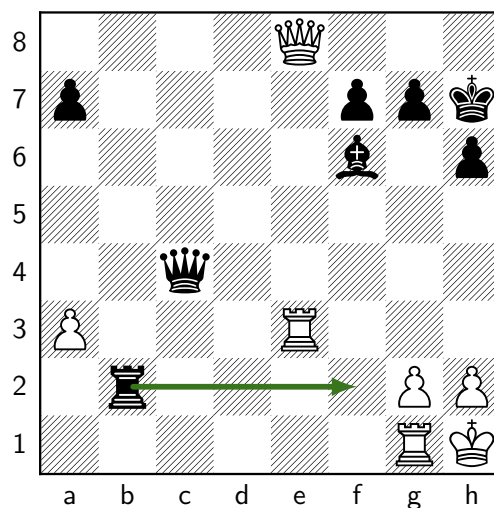
(a)



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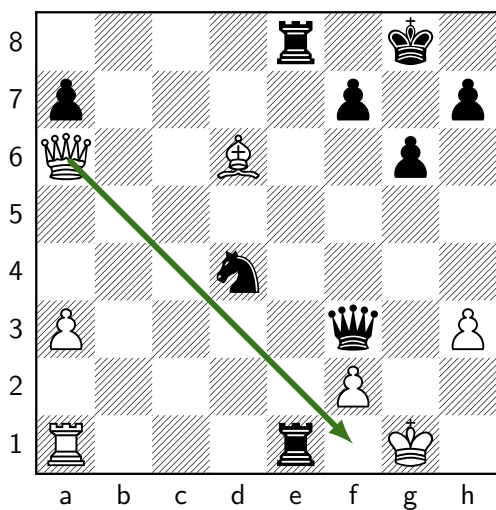


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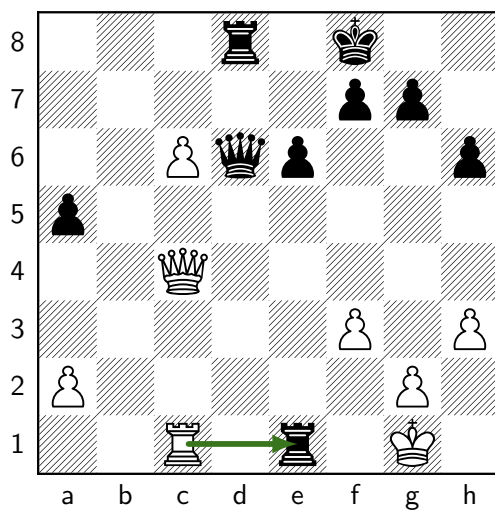


(d)

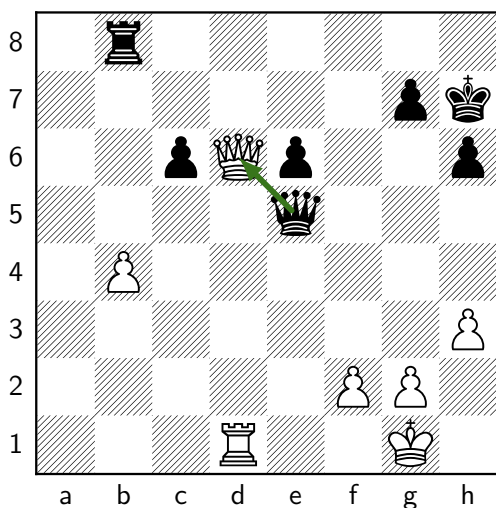
Figure 18: **Move 25, White: h3.** (a) Our position; Melanie plays h3 (h2h3). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0000#2926` (ply 60) [structural distance $d=53$, QAP obj=40]: the master played **Bxg6**; under μ our *Qa6* plays the role of that game’s *Bd3* (a **cross-type** correspondence); cross-type roles: our *Bd6* as their *Pg2*, our *nd4* as their *pd6*; transferred move **a6a7**. (c) Retrieved from `ccr1_top_p0000#78` (ply 61) [structural distance $d=55$, QAP obj=38]: the master played **Ra7**; under μ our *Qa6* plays the role of that game’s *Rd2* (a **cross-type** correspondence); cross-type roles: our *Ra1* as their *Qb8*, our *Ph2* as their *Bf8*; transferred move **a6a5**. (d) Retrieved from `ccr1_top_p0000#3995` (ply 63) [structural distance $d=52$, QAP obj=22]: the master played **Rf2**; under μ our *Ph2* plays the role of that game’s *Rb7* (a **cross-type** correspondence); cross-type roles: our *Bd6* as their *Pg7*, our *nd4* as their *Bf6*; transferred move **h2h4**. Synthesis: no retrieved analog transferred to **h3** – the **search** selector chose it on value (material / mate); panels (b)–(d) show the strongest nearby analogies.



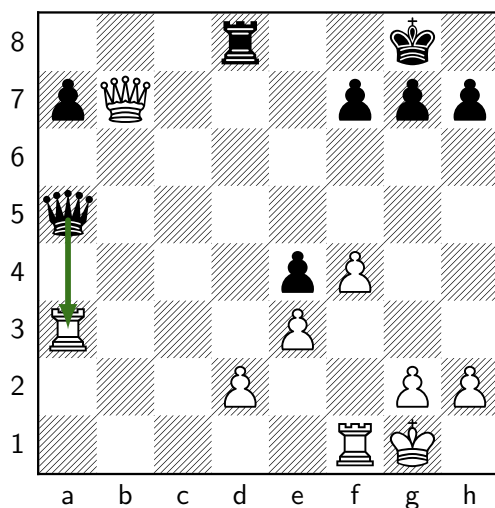
(a)



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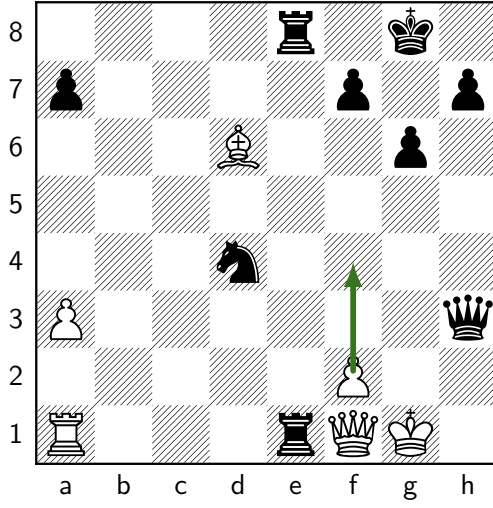


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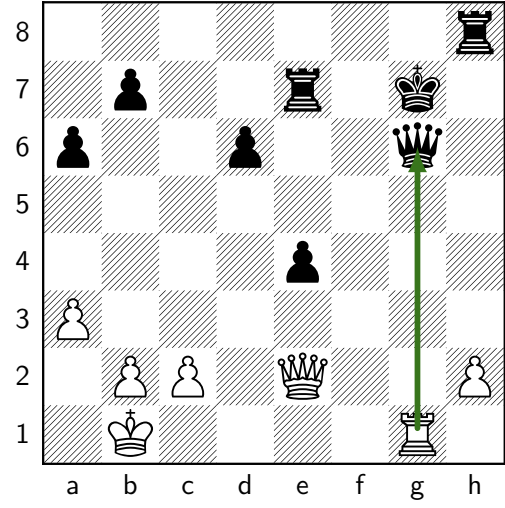


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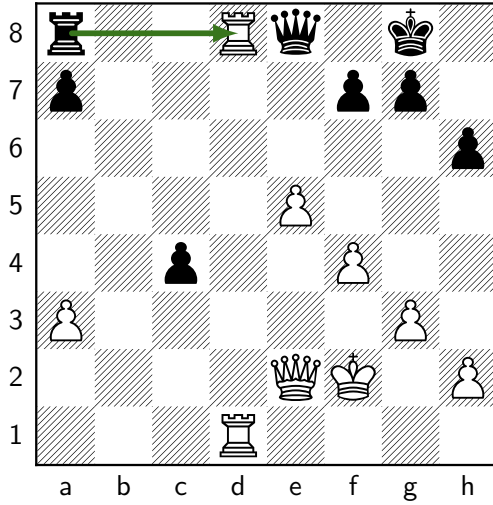
Figure 19: **Move 26, White: Qf1**. (a) Our position; Melanie plays Qf1 (a6f1). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0002#3030` (ply 64) [structural distance $d=57$, QAP obj=69]: the master played Rxe1; under μ our Qa6 plays the role of that game’s Rc1 (a **cross-type** correspondence); cross-type roles: our Bd6 as their Pg2, our Ra1 as their Qc4; transferred move a6f1 (**endorses this move**). (c) Retrieved from `ccr1_top_p0003#2677` (ply 67) [structural distance $d=57$, QAP obj=4]: the master played Qxd6; under μ our Qa6 plays the role of that game’s Qe4; cross-type roles: our Bd6 as their Pc6, our re1 as their qd6; transferred move a6f1 (**endorses this move**). (d) Retrieved from `ccr1_top_p0001#2336` (ply 45) [structural distance $d=56$, QAP obj=3]: the master played Qxa3; under μ our Qa6 plays the role of that game’s Qa4; cross-type roles: our pg6 as their Rd8, our Ra1 as their Pf7; transferred move a6f1 (**endorses this move**). Synthesis: 3 retrieved analogs transferred to Qf1 (summed QAP objective 76, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



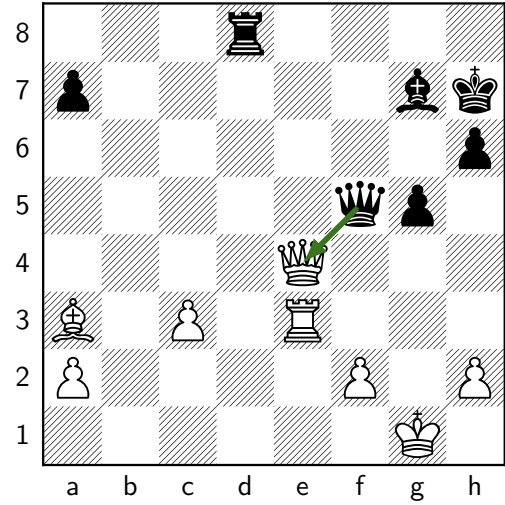
(a)



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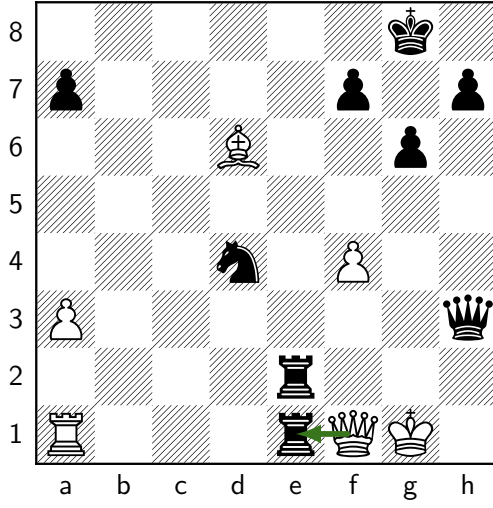


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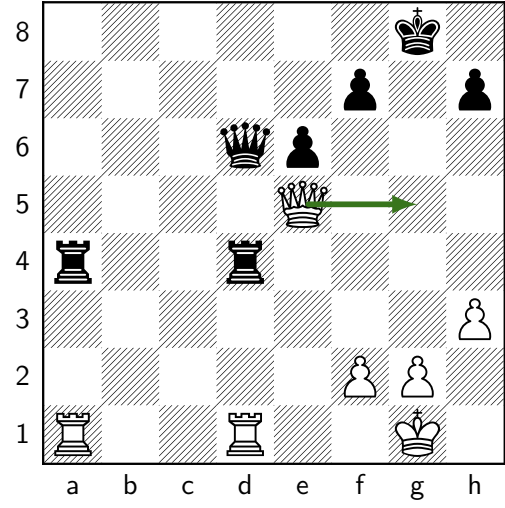


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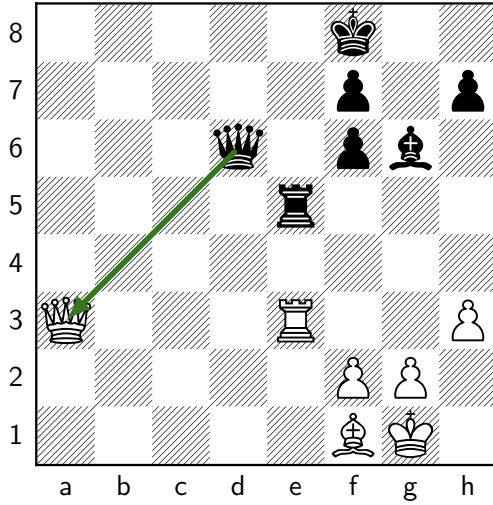
Figure 20: **Move 27, White: f4.** (a) Our position; Melanie plays f4 (f2f4). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccrl_top_p0003#1283` (ply 68) [structural distance $d=63$, QAP obj=4]: the master played $Rxg6+$; under μ our $Pf2$ plays the role of that game’s $Rg1$ (a **cross-type** correspondence); cross-type roles: our $Bd6$ as their $Pb2$, our $qh3$ as their $Pc2$; transferred move $f2f4$ (**endorses this move**). (c) Retrieved from `ccrl_top_p0002#2172` (ply 55) [structural distance $d=60$, QAP obj=60]: the master played $Rxd8$; under μ our $Ra1$ plays the role of that game’s $Ra1$; cross-type roles: our $Bd6$ as their $Pg7$, our $nd4$ as their $Ph6$; transferred move $a1e1$. (d) Retrieved from `ccrl_top_p0003#1444` (ply 63) [structural distance $d=67$, QAP obj=57]: the master played $Qxe4$; under μ our $Qf1$ plays the role of that game’s $Qf4$; cross-type roles: our $pg6$ as their $Rd8$, our $Ra1$ as their $Pa7$; transferred move $f1e1$. Synthesis: 1 retrieved analog transferred to f4 (summed QAP objective 4, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.



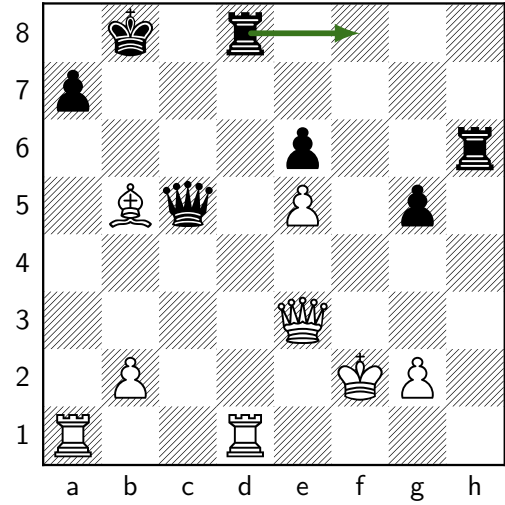
(a)



(b)



(c)



(d)

Figure 21: **Move 28, White: Qxe1.** (a) Our position; Melanie plays **Qxe1** (**f1e1**). Panels (b)–(d) are the retrieved master positions whose correspondence most drove this move (highest QAP objective among those that transfer to it). (b) Retrieved from `ccr1_top_p0003#1723` (ply 54) [structural distance $d=81$, QAP obj=56]: the master played **Qg5+**; under μ our *Qf1* plays the role of that game's *Qe5*; cross-type roles: our *pg6* as their *Ra1*, our *qh3* as their *Pg2*; transferred move **f1e1** (**endorses this move**). (c) Retrieved from `ccr1_top_p0000#2874` (ply 71) [structural distance $d=88$, QAP obj=30]: the master played **Qxa3**; under μ our *Qf1* plays the role of that game's *Qd3*; cross-type roles: our *re2* as their *Pf6*, our *pg6* as their *Bg6*; transferred move **f1e1** (**endorses this move**). (d) Retrieved from `ccr1_top_p0002#3203` (ply 59) [structural distance $d=78$, QAP obj=34]: the master played **Rf8+**; under μ our *Pa3* plays the role of that game's *Rd1* (a **cross-type** correspondence); cross-type roles: our *Bd6* as their *Pa7*, our *nd4* as their *Pe6*; transferred move **a3a4**. Synthesis: 2 retrieved analogs transferred to **Qxe1** (summed QAP objective 86, the strongest correspondence among the candidate moves); Melanie committed it via the **search** selector.